

# VisionVerse 3D Commerce

"Shop Beyond the Screen."

An immersive spatial e-commerce ecosystem integrating real-time Three.js/WebGL procedural 3D models, hardware telemetry hotspots, live PBR customizer, WebXR augmented reality, and persistent client-side checkout.

**Stack:** HTML5 · CSS3 Glassmorphism · Vanilla JS · Three.js ·  
WebGL · LocalStorage

**GitHub:** [github.com/nirmalameka6/3D-Websites-Hackathon](https://github.com/nirmalameka6/3D-Websites-Hackathon)

## 1. Executive Summary

**VisionVerse 3D Commerce** is a complete, zero-backend spatial e-commerce web platform designed to eliminate flat, static 2D product photos. By utilizing high-performance WebGL and procedural Three.js rendering, customers can interact with products in 360 degrees, inspect internal engineering with a  **3D Explode View**, customize materials and colors live with dynamic pricing, project items into their space with **Augmented Reality (AR)**, and checkout seamlessly with LocalStorage state persistence.

## 2. Problem Statement vs. VisionVerse Solution

| Traditional 2D E-Commerce                                                                    | VisionVerse 3D Spatial Commerce                                                                      |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Flat 2D Photos:</b> Shoppers guess how items look in real life; high return rates (30%+). | <b>360° Real-Time 3D:</b> Orbital rotation, smooth zoom, and physically based rendering (PBR).       |
| <b>Static Color Swatches:</b> Customers can't see realistic material/finish behavior.        | <b>Live 3D Customizer:</b> Switch between Titanium, Glass, Carbon, and Polymers in real-time.        |
| <b>Text Walls for Specs:</b> Shoppers skim through specs without understanding hardware.     | <b>Interactive Hotspots:</b> Clickable 3D pins projecting component telemetry directly on the model. |
| <b>Passive Browsing:</b> Standard grids feel like reading a static digital magazine.         | <b>Virtual 3D Exhibition:</b> Walkable 3D digital showroom with cinematic camera gliding.            |

### 3. Technical Architecture & Modular Design

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visionverse-3d-commerce/ |— index.html # Hero 3D Background, Micro-Tech Grid, 3D Showcase
& Virtual Exhibition |— shop.html # Full 3D Catalog with Live Search, Price Slider (₹10 -
₹10L), and Filters |— product.html # Master 3D Product Studio, Hotspots, Live Customizer,
Explode View, AR |— cart.html # Shopping Cart with Customization Tags, Steppers, and
Promo Coupons |— checkout.html # Express Checkout, Interactive Credit Card Widget, UPI
QR, Confirmation | |— js/ | |— products.js # Unified database of 20+ products (₹19 to
₹8,99,999) | |— three-viewer.js # Procedural Three.js 3D Engine, Studio Lights, Explode
Mode, Particles | |— main.js # Shared state (Cart/Wishlist), Toast system, 3D Hero canvas
| |— shop.js # Live search, dynamic category filtering, and card rendering | |—
product.js # Product detail controller, Studio lights, Explode mode, Dynamic pricing | |—
cart.js # Cart quantity stepper, discount promo engine, shipping threshold | |—
checkout.js # Demo payment validation, live card preview, order receipt generation | |—
ai-guide.js # Rule-based Vision AI assistant with budget query NLP |— css/ |— style.css
# Core theme, glassmorphism, navigation, AI widget, and modals |— shop.css # Catalog
grid, filter sidebar, and card hover animations |— product.css # 3D viewport, studio
lighting chips, explode button, and customizer |— cart.css # Responsive cart table,
stepper, and summary box |— checkout.css # Checkout forms, live credit card widget, and
confirmation modal

```

## 4. Product Catalog: From ₹19 Micro-Tech to Hyper-EVs

The platform features 20+ procedural 3D products spanning budget micro-accessories up to hyper-cars:

| Product                   | Category    | Price (INR) | 3D Features & Highlights                                         |
|---------------------------|-------------|-------------|------------------------------------------------------------------|
| Cyber NFC Smart Tag       | Accessories | ₹19         | Rotating holographic diffractive film & etched micro-IC chip     |
| Quantum Lens Guard        | Accessories | ₹49         | Triple 9H sapphire optical ring assembly with refractive glass   |
| Cyber Beacon Keyring      | Accessories | ₹99         | Levitating crystal with dual counter-rotating gyro rings         |
| Acoustic Ear Tips         | Audio       | ₹149        | Viscoelastic memory foam buds with acoustic core tubes           |
| Braided RGB Warp Cable    | Accessories | ₹199        | Flowing photon waveguide with USB-C 240W zinc alloy head         |
| Cyber Stylus Pen          | Accessories | ₹299        | Magnetic inductive charging strip, 4096-level POM fine nib       |
| MagSafe Cyber Puck        | Accessories | ₹499        | 360° pulsating neon breathing halo & Qi2 copper coil             |
| Desk Hologram Prism       | Smart Home  | ₹899        | Levitating K9 crystal pyramid with sound-reactive RGB beams      |
| Cyber Fitness Band        | Wearables   | ₹999        | Curved vertical OLED screen, 24/7 PPG bio-telemetry              |
| Vision GamePad Controller | Gaming      | ₹1,999      | Hall-Effect anti-drift joysticks, RGB edge grips, micro-triggers |
| VisionPods Headphones     | Audio       | ₹3,499      | Graphene drivers, -48dB hybrid ANC, spatial 3D audio             |

| Product             | Category    | Price (INR) | 3D Features & Highlights                                    |
|---------------------|-------------|-------------|-------------------------------------------------------------|
| Smart Ring          | Wearables   | ₹4,499      | Medical-grade titanium band, continuous bio-impedance       |
| VisionWatch X1      | Wearables   | ₹4,999      | 1.96" curved AMOLED, titanium matrix, haptic crown dial     |
| Smart Home Hub      | Smart Home  | ₹5,999      | 360° LiDAR presence scanner, levitating hologram sphere     |
| Smart Speaker       | Audio       | ₹6,999      | Downward neodymium subwoofer, 32-zone aura top ring         |
| VisionWatch Pro     | Wearables   | ₹7,999      | Solar sapphire glass, dual-frequency GNSS, 100m dive rating |
| Vision Glasses (AR) | Wearables   | ₹12,999     | Micro-OLED diffractive waveguides, bone conduction audio    |
| Vision Camera       | Optics      | ₹24,999     | Full-frame 8K raw sensor, 5-axis OIS, tactile dials         |
| Future Drone        | Drones      | ₹34,999     | 4K gimbal eye, omnidirectional LiDAR radar, spinning rotors |
| VisionPhone X       | Smartphones | ₹49,999     | 200MP periscope array, 3nm neural core, titanium unibody    |
| VisionLaptop Pro    | Computing   | ₹79,999     | 16" 4K OLED 165Hz screen, liquid gallium vapor cooling      |
| VisionCar One       | Vehicles    | ₹8,99,999   | Level 4 autonomous AI coupe, solid-state 950km battery      |

## 5. Core 3D Spatial Innovations

### 1. Real-Time Hardware Hotspots

Uses mathematical 3D-to-2D screen coordinate projection in the animation loop: projects Three.js vector (x, y, z) into screen coordinates (px, py), rendering pulsing DOM tooltips that follow the

model during rotation.

## 2. 🌟 3D Explode / Deconstruct View

Smoothly animates components outward along their normal vectors, letting users inspect internal engineering layers and sensor coils.

## 3. 💡 Multi-Environment Studio Lighting

Live switching between **Cyber Neon** (Cyan/Magenta), **Clean Studio** (5500K White), **Sunset Amber** (Warm Gold), and **Matrix Glow** (Emerald).

## 4. 🎨 Live Customizer & Dynamic Pricing Engine

Real-time PBR material shader switches (Titanium, Glass, Carbon, Polymer) with instantaneous recalculation:  $\text{Total} = \text{Base Price} + \text{Material Extra} + \text{Size Config Extra}$ . Customizations are encapsulated into unique cart signatures.

## 5. 🕶️ WebXR & Holographic AR Mode

Detects native WebXR support with fallback to an interactive mobile QR code generator and scale placement simulator.

## 6. 🛒 Vision AI Shopping Guide

Smart rule-based NLP assistant that answers budget queries ("Show items under ₹100"), compares hardware ("Compare X1 vs Pro"), and directly manipulates the 3D viewport.

# 6. Hackathon Presentation Script (Ready to Deliver)

*"Hi judges! Traditional e-commerce is stuck on flat, 2D pages. VisionVerse 3D Commerce brings shopping to life with our tagline: '**Shop Beyond the Screen.**'"*

*Built with Three.js, WebGL, and Vanilla JavaScript, VisionVerse hosts a catalog of 20+ procedural 3D products from ₹19 micro-gadgets to flagship hyper-EVs. Customers can rotate products in 360 degrees, click hardware hotspots, trigger 3D Explode Views to inspect internal engineering, customize materials live with real-time dynamic pricing, and project products in Augmented Reality.*

*With an interactive 3D virtual exhibition, a smart Vision AI assistant, and a full checkout*

*pipeline with persistent local storage, VisionVerse bridges the physical-digital divide and redefines the future of online shopping. Thank you!"*

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[github.com/nirmalameka6/3D-Websites-Hackathon](https://github.com/nirmalameka6/3D-Websites-Hackathon)